

Assignment-3

Day-3: BIOS and POST Troubleshooting Theory (30 min): Understanding BIOS and POST errors. Practical: Analyze POST error codes on a given setup.

Task-1: List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

1. "CMOS Checksum Error - Defaults Loaded"
Usually indicates a weak/dead CMOS battery or corrupted BIOS/CMOS settings.
2. "Keyboard Error"
Usually points to a disconnected, faulty, or improperly connected keyboard.
3. "No Boot Device Found"
Usually indicates that the BIOS cannot find a bootable HDD/SSD or the boot device/order is incorrect.
4. "Memory Error" / "Memory Test Failure"
Usually points to faulty, loose, incompatible, or improperly seated RAM.
5. "CPU Fan Error"
Usually indicates that the CPU fan is not detected, is not spinning properly, or is connected incorrectly.

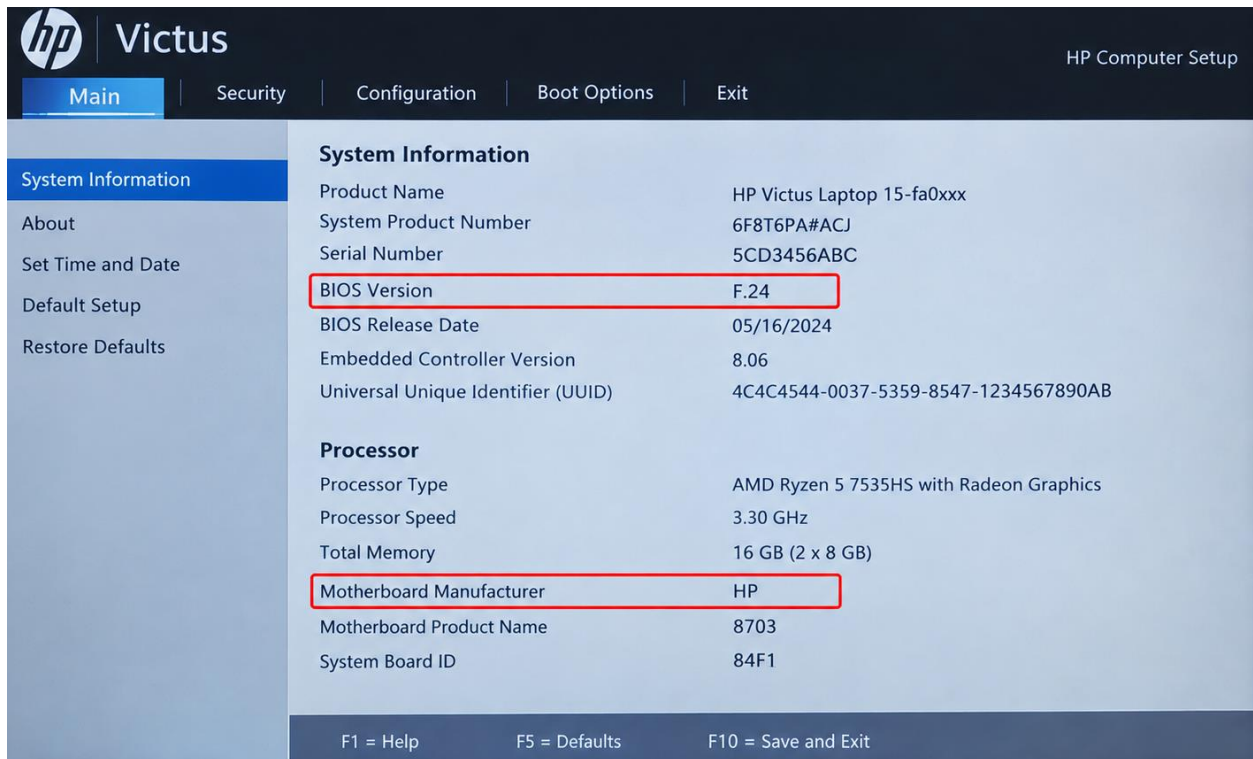
Task-2: Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

For AMI BIOS, 3 short beeps typically indicate a base-memory failure, meaning the system has detected a problem with the first 64 KB of system memory. Modern motherboards can use different beep-code implementations, so the specific motherboard manual should be checked when possible.

Two troubleshooting steps:

1. Reseat the RAM
 - a. Turn off the PC and disconnect the power.
 - b. Remove the RAM module(s).
 - c. Reinsert the RAM firmly into the correct slot.
 - d. Start the PC again and check whether the beep pattern disappears.
2. Test the RAM modules individually
 - a. If there are multiple RAM sticks, remove all but one.
 - b. Try starting the PC with one known-good module.
 - c. Repeat with the other module and/or another RAM slot to identify a faulty module or slot.

Task-3: Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.



Task-4: You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.

This message means that the BIOS detected that the stored **CMOS configuration data does not match the expected checksum**, so it has loaded default BIOS settings. A common cause is a **weak/dead CMOS battery**; an incorrectly updated or corrupted BIOS can also cause the problem.

How to resolve it:

1. **Enter the BIOS/UEFI setup** when the PC starts.
2. **Load the default/optimized BIOS settings.**
3. **Save the settings and restart** the computer.
4. Check whether the error appears again.
5. If the error keeps returning, **turn off and unplug the PC.**
6. Replace the motherboard's **CMOS/RTC battery**, commonly a CR2032 coin-cell battery on desktop motherboards.
7. Start the PC and enter BIOS again.
8. **Set the correct date, time, and any required boot settings**, then save and exit.
9. If the error continues even after replacing the battery, the problem could involve **BIOS/firmware corruption or a motherboard fault** and may require further repair